



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 02

Experiment Title: Design and Simulation of an 8-Bit Full Adder

Submitted by:

Name: Naima Tabassum Antora

Student ID: 22201068

Section: C1

Submitted to:

Md. Rasheduzzaman

Lecturer

Department of Computer Science and Engineering

Date of Submission: 27 September, 2026

1. Experiment Name

Design and Simulation of an 8-Bit Full Adder

2. Objective

- To design and implement an 8-bit Full Adder using Verilog HDL.
- To build the 8-bit adder by connecting multiple 1-bit Full Adder stages so that the carry from one stage is passed to the next stage.
- To synthesize the design in Xilinx ISE 14.7 and observe its RTL schematic.
- To develop a testbench and simulate the design using ISim with different input combinations.
- To verify that the simulated sum and carry outputs match the expected binary addition results.

3. Apparatus / Hardware & Software Requirements

- **Xilinx ISE 14.7** — used to write the Verilog code, synthesize the design, and generate the RTL schematic.
- **ISim Simulator** — used to run the testbench and observe the input and output waveforms.
- **Verilog HDL** — used to describe both the 1-bit Full Adder module and the complete 8-bit Full Adder design.
- **Oracle VirtualBox** — used to run the Xilinx ISE virtual machine environment.

4. Circuit Diagram / Schematic

The 8-bit Full Adder was built by connecting eight 1-bit Full Adder stages. The carry output of each stage is passed to the next stage. The design uses separate inputs a0–a7, b0–b7, and cin, and produces sum0–sum7 with a final carry output. The RTL schematic was generated in Xilinx ISE 14.7 to observe the overall connection of the design.

Figure 1

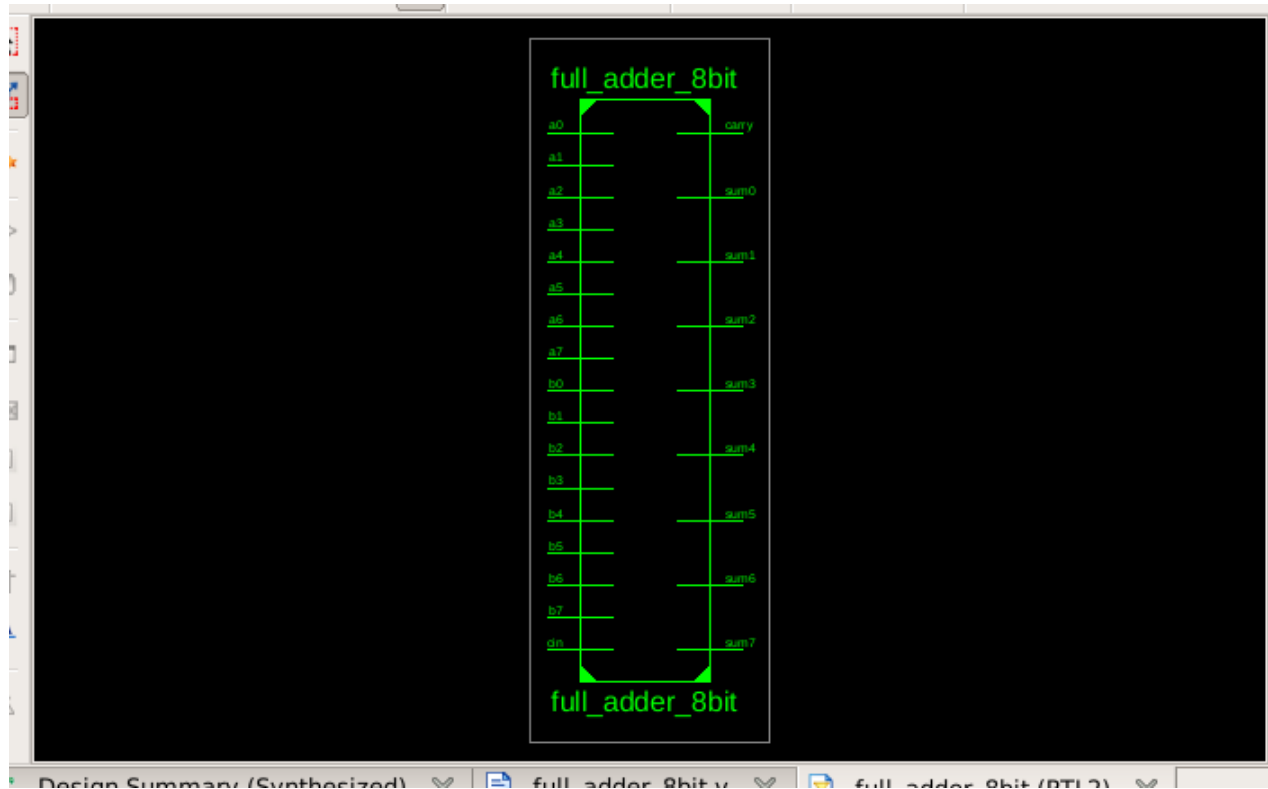


Figure 1: Top-Level RTL Schematic of the 8-Bit Full Adder

Figure 1 shows the top-level RTL view of the 8-bit Full Adder. The input and output signals are shown separately, including the eight input bits for each operand, the carry input, the eight sum outputs, and the final carry output.

Figure 2

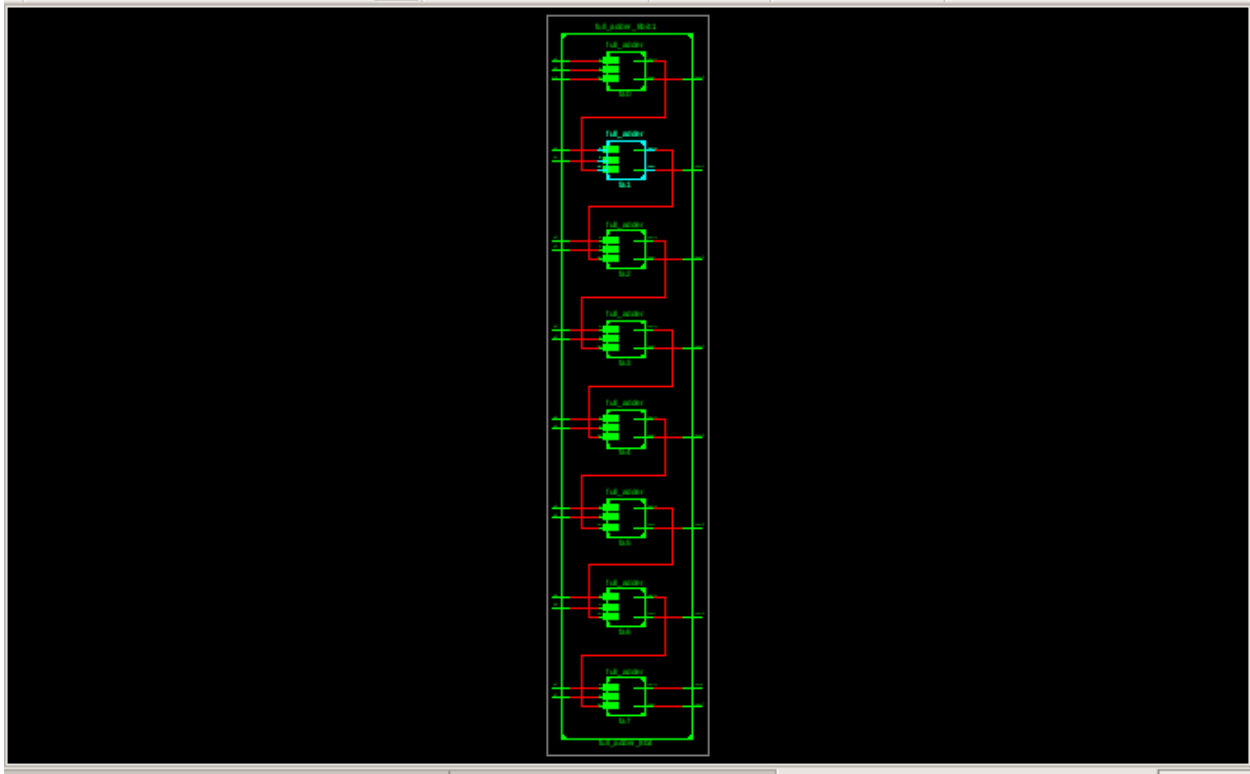


Figure 2: Expanded RTL Schematic of the 8-Bit Full Adder

Figure 2 shows the internal structure of the design, where eight 1-bit Full Adder blocks are connected together. The carry generated by each Full Adder is passed to the next stage, allowing the complete 8-bit binary addition to be performed.

6. Output Analysis / Observations

The 8-bit Full Adder was tested in ISim using several input combinations to check whether the sum and carry outputs were produced correctly. Each test case was applied for 100 ns, and the waveform was observed throughout the simulation.

The simulation included simple additions as well as cases where the carry had to pass through several Full Adder stages. The observed outputs matched the expected binary addition results in each case.

Table 1: 8-bit Full Adder input addition and expected result

Test Case	Input Addition	Expected Result
1	0 + 0	Sum = 0, Carry = 0
2	0 + 0 + Cin(1)	Sum = 1, Carry = 0
3	1 + 1	Sum = 2, Carry = 0

Test Case	Input Addition	Expected Result
4	5 + 3	Sum = 8, Carry = 0
5	15 + 1	Sum = 16, Carry = 0
6	127 + 1	Sum = 128, Carry = 0
7	255 + 1	Sum = 0, Carry = 1

The last test case was especially useful for checking the final carry output. When **255 + 1** was applied, all eight sum outputs became 0 and the final **carry** became 1, which represents the correct 9-bit result of 256.

Figure 3

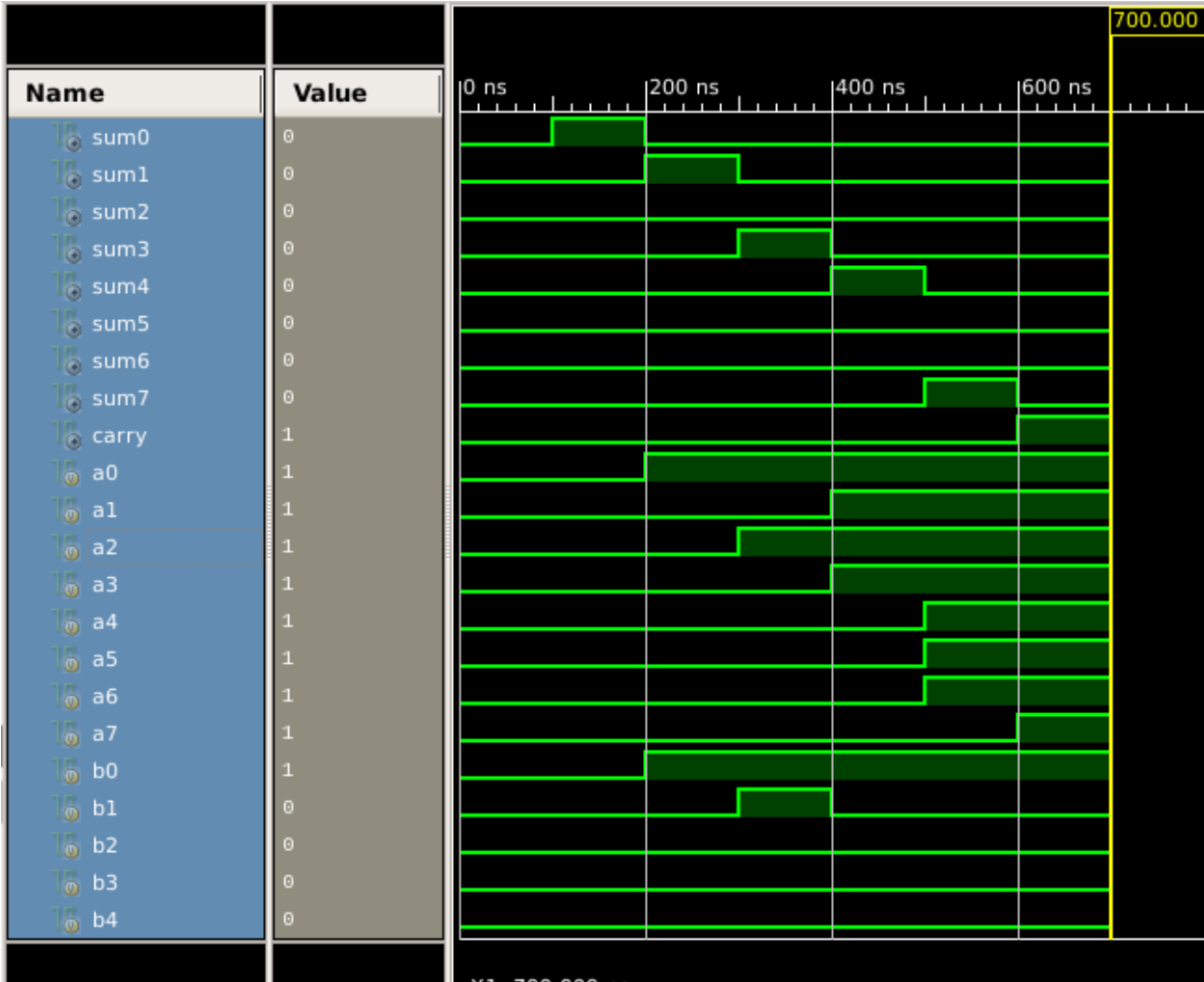


Figure 3: Simulation Waveforms of Input Signals of the 8-Bit Full Addder (a)

Figure 4

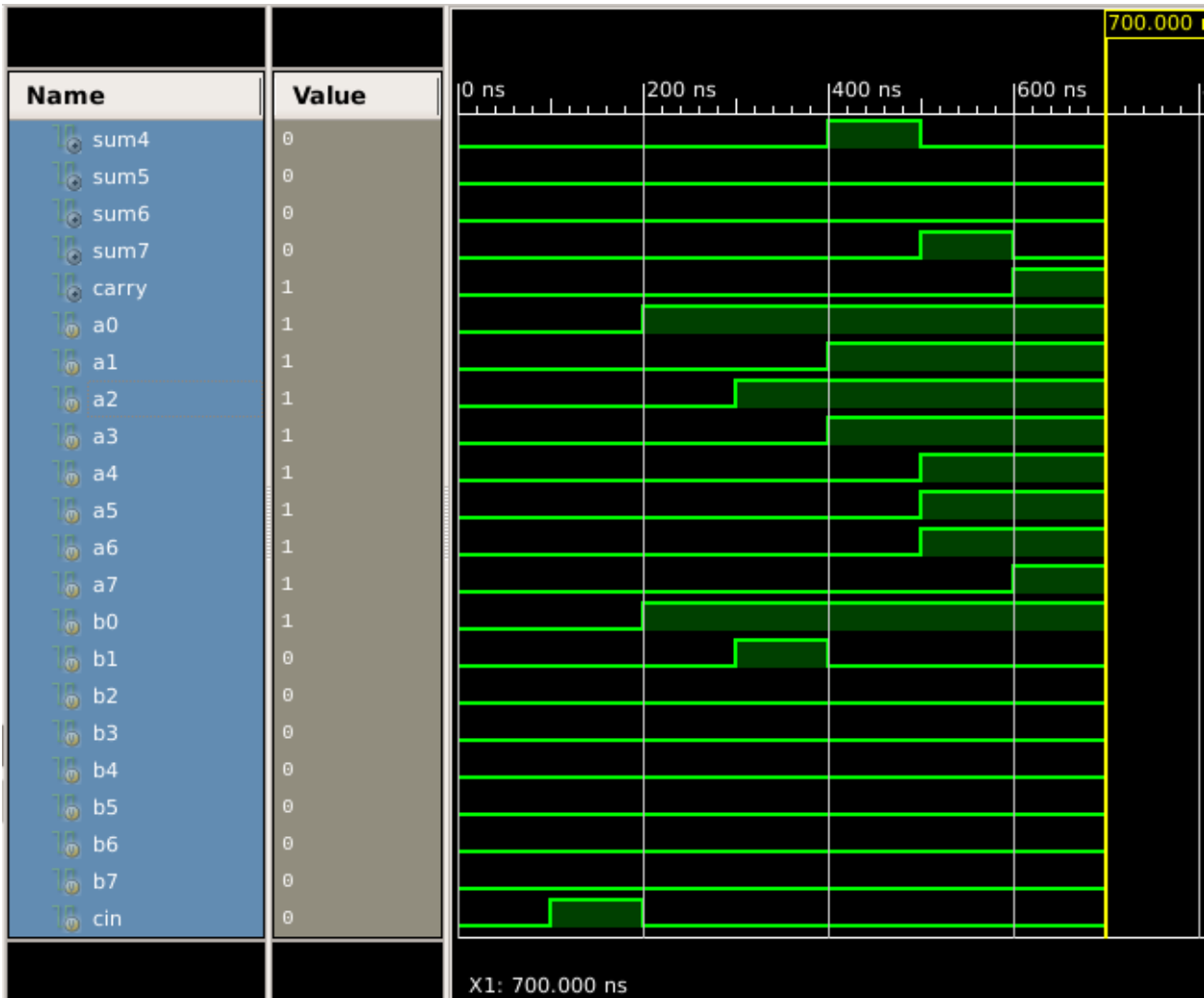


Figure 4: Simulation Waveforms of Input Signals of the 8-Bit Full Adder (b)

Table 2: Simulation Observation of the 8-bit Full Adder

Time (ns)	A7–A0	B7–B0	Cin	Sum7–Sum0	Carry
0–100	00000000	00000000	0	00000000	0
100–200	00000000	00000000	1	00000001	0

Time (ns)	A7–A0	B7–B0	Cin	Sum7–Sum0	Carry
200–300	00000001	00000001	0	00000010	0
300–400	00000101	00000011	0	00001000	0
400–500	00001111	00000001	0	00010000	0
500–600	01111111	00000001	0	10000000	0
600–700	11111111	00000001	0	00000000	1

The observed Sum and Carry outputs matched the expected results for all applied input combinations.

7. Result

The 8-bit Full Adder was successfully designed and implemented using Verilog HDL in Xilinx ISE 14.7. The design was synthesized correctly, and the RTL schematic showed the connection of eight 1-bit Full Adder stages. The testbench was simulated in ISim using several input combinations, and the observed **sum** and **carry** outputs matched the expected binary addition results. The final carry condition was also verified with the **255 + 1** test case. Therefore, the main objective of the experiment was achieved successfully.

8. Github Repository Commits

Commit: Add files via upload

Commit Link: <https://github.com/naima-tabassum/CSE450-VLSI-Lab/commit/f6c301977aa0620b4179aa52cdd6d87f149c3a16>

9. Conclusion

This experiment helped me understand how a larger digital circuit can be built by connecting several 1-bit Full Adder stages together. I learned how the carry from one stage moves to the next stage and how this allows an 8-bit binary addition to be completed. I also gained practical experience in writing the Verilog HDL design, generating the RTL schematic in Xilinx ISE 14.7, creating a testbench, and checking the results in ISim. By comparing the simulated sum and carry outputs with the expected binary addition results, I was able to verify that the design worked

correctly. Overall, this experiment improved my understanding of multi-bit adder design, carry propagation, and simulation-based verification.